

SE-LAB1

Scenerio 45:

Req-ID	Type	Description	Prio rity	Acceptance Criteria	Rationale
FR-001	Function al	The system shall ingest cloud billing CSV exports, categorize costs by service and project tag, and highlight compute instances with < 5% CPU utilization over 14 days	Hig h	Pass: Idle cloud asset flagged with monthly savings estimate; Fail: Active instance misclassified as idle.	Establishes the core cost-visibility function the portal is built around.
FR-002	Function al	The system shall scan storage inventories across all connected cloud accounts and flag storage volumes that have been unattached to any compute instance for more than 7 consecutive days.	Hig h	Pass: An unattached volume idle for 7+ days appears in the Idle Assets list with size and monthly cost. Fail: An attached volume is listed as unattached.	Unattached storage is a common, easily-automated source of wasted spend.
FR-003	Function al	The system shall dispatch a budget breach alert (email and in-app notification) to the Cloud Administrator when actual monthly spend for a project exceeds the configured budget threshold.	Hig h	Pass: Alert is delivered within 5 minutes of the breach being detected. Fail: No alert sent, or sent to the wrong recipient.	Timely alerts let administrators act before overspend compounds.
FR-004	Function al	The system shall allow a FinOps Analyst to configure and edit a monthly budget threshold per project or cost center.	Me diu m	Pass: Updated threshold is saved and used for the next breach check. Fail: Threshold edit is not persisted or not applied.	Budgets vary by project; thresholds must be analyst-editable, not hard-coded.
FR-005	Function al	The system shall generate a downloadable cost-optimization report (PDF/CSV) summarizing idle assets, unattached volumes, and estimated monthly savings for a selected date range.	Me diu m	Pass: Exported report matches on-screen totals for the selected range. Fail: Report is missing assets shown on screen or totals mismatch.	Analysts need an exportable artifact for stakeholder reviews.

NFR-001	Non Functional [Type: Performance & Security]	The portal must parse multi-gigabyte cloud billing files and update cost dashboards in under 2 minutes.	High	Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.	Large billing exports are routine; slow ingestion or weak transport security blocks adoption.
NFR-002	Non Functional	The system shall enforce role-based access control (RBAC) so that only users with the Cloud Administrator role can modify budget thresholds or dismiss breach alerts, and all billing data must be encrypted at rest (AES-256).	High	Pass: A FinOps Analyst account attempting to edit a threshold is denied with an authorization error; data-at-rest encryption is confirmed via storage audit. Fail: Any unauthorized role successfully modifies a threshold, or data is found unencrypted.	Billing data is sensitive financial data; access must be restricted and protected at rest.

UML DIAGRAM



Primary Actor:

- FinOps Analyst (triggers/reviews analysis)
- Cloud Administrator (receives alert)

Preconditions:

- FinOps Analyst is authenticated (Login is included by this use case).
- At least one valid AWS, Azure, or GCP billing CSV has been uploaded to the portal.
- A budget threshold has been configured for the project being analyzed.

Postconditions:

- Idle compute instances and unattached storage volumes are flagged with estimated monthly savings.
- If a budget breach is detected, the Cloud Administrator has received an alert (email + in-app).
- An audit log entry records the analysis run and any alert dispatched.

Main Success Scenario:

1. FinOps Analyst logs in and selects "Upload Billing CSV". System validates and ingests the file.
2. System categorizes costs by service and project tag.
3. System scans compute instances for < 5% CPU utilization over the last 14 days and storage volumes unattached for 7+ days.
4. System flags each idle asset with an estimated monthly savings figure and displays them in the Idle Assets dashboard.
5. System compares current project spend against the configured budget threshold (extends this use case with "Apply Custom Threshold Rules" if the analyst has defined project-specific rules).
6. If spend exceeds the threshold, system generates a budget breach alert.
7. System dispatches the alert to the Cloud Administrator via email and in-app notification.
8. Cloud Administrator reviews the alert and the flagged idle assets.
9. Use case ends successfully.

Alternate Flow — 1a. Corrupted or Invalid CSV File:

1. System detects the uploaded file fails schema validation (missing columns, unreadable encoding, etc.).
2. System rejects the file and displays an error message identifying the specific validation failure.
3. FinOps Analyst corrects the file and re-uploads it, or contacts the Cloud Administrator if the issue persists.
4. If re-upload succeeds, flow resumes at Main Success Scenario step 2; otherwise the use case ends in failure.